



A Novel Approach for Better Career Counselling Utilizing Machine Learning Techniques

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Abstract

The biggest issue many students face in today's world is choosing the right career. Especially when there are so many options available to them and the counselling options available are very limited or not very efficient, Career counselling is a very essential process that assists individuals in making informed decisions about their career paths. The use of machine learning in career counselling has gained so much attention due to its potential to analyse vast amounts of data and provide personalised guidance. Previously, there had been so much work done in this field with the help of artificial intelligence and machine learning, but there was a lack of a systematic system where students could explore each and every option thoroughly and get to know what the real outcome would be if they chose that stream. In this study, various factors such as the student's interests, hobbies, past academics and performances, and achievements are taken into consideration to predict the right career option. The model is trained using five different machine learning algorithms: decision tree, Random Forest, Support Vector Machine, Nave Bayes, and K-nearest neighbours Classifier. Out of these, Random Forest gave the highest accuracy of 84.17%, and after hypertuning, it gave the highest accuracy of 85.68%. We also gave some manual inputs to the system and found out that the Random Forest gave the highest accuracy of 85.71%. The prediction results of each algorithm are summarised in this study.

Keywords Machine learning · Decision tree · Random forest · Support vector machine (SVM) · Naïve Bayes · KNN

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Abbreviations

AI	Artificial intelligence
ANN	Artificial neural networks
C3-IOC	AI-based solution
CSS	Cascading style sheets
ES	Expert system
F1-Score	Harmonic mean of precision and recall
FN	False negative
FP	False positive
GridSearchCV	GridSearch cross-validation
HTML	Hypertext markup language
ICT	Information and communication technology
IT	Information technology
KNN	K-nearest neighbors algorithm
MySQL	My structured query language
NB	Naive Bayes
PHP	Hypertext preprocessor
RBF	Radial basis function
SKlearn	Scikit-learn
SVM	Support vector machine
TN	True negative
TP	True positive
XGBoost	Extreme gradient boosting

1 Introduction

Education counselling is becoming increasingly important in the field of education as the field of education is growing more complex. With so many options and pathways available to students, it can be very difficult to navigate the educational landscape without the guidance and support of a trained professional [1]. From selecting a course of study to exploring career opportunities and preparing for the workforce, education Counsellor plays a huge role. By helping students to clarify their goals and develop a clear plan for achieving them, education counselling can make a significant difference in the success and satisfaction of students throughout their academic and professional careers. The use of technology has revolutionized the way we approach career counselling.

With the increasing Technology, career counsellors can now provide more comprehensive and accessible support to their clients online. One of the major benefits of technology in career counselling is the potential to deliver personalized guidance and advice to students on a large scale. Online assessments and career exploration tools help Students to identify their strengths and interests and provide them with information on potential career paths that align with their skills and passions.

Overall, the use of technology in career counselling has the power to make this important service more mobile, efficient, and effective for students at all stages of their career development [2].

In a country like India, Students face a lot of challenges while choosing a career option. In India, we can see a lot of coaching centers, a lot of good schools and colleges, and a lot of startups in education but hardly see something which helps students to know and select

the right career option. A student generally chooses career options based on parental or societal pressure or goes with what their friends are taking. Addressing these challenges will require a comprehensive approach that includes providing access to information, promoting career counselling services, encouraging a diverse range of career options, and reducing the influence of societal pressure on career decisions.

In the field of career counselling, so much work had been already done. We all can see the wide use of machine learning and artificial intelligence which take the information of a student such as records, current interests, and suggest them a career option without even knowing that the student even knows or not about that field. If we see the internet, is flooded with immense information, if a student acquires knowledge from there, it will eventually lead to confusion, and the internet has different people giving different information on a particular topic. Therefore, judging all these issues, we came up with a study that will help the career counselling domain. Machine Learning has the potential to revolutionize career counselling by providing data-driven insights, personalized recommendations, and predictive analytics and seeing all this, we developed a machine learning model using five different machine learning algorithms and a detailed study is summarized in this paper.

- 1.1. *Motivation*: Employing classification machine learning techniques in career counseling offers the potential for data-driven, personalized guidance, efficient and scalable services, improved accuracy, access to diverse career options, and continuous learning and improvement. By harnessing the power of technology and data, these techniques can enhance the effectiveness of career counseling and support individuals in making informed decisions about their professional paths.
- 1.2. *Novelty of the work*: Integration of machine learning: The novel aspect is the application of machine learning techniques to the field of career counseling. This integration involves using classification algorithms or other machine learning approaches to improve the accuracy and effectiveness of career counseling recommendations.

Dataset selection: The work introduces a unique or comprehensive dataset specifically tailored for career counseling purposes. This dataset contains various attributes related to individuals' skills, interests, educational background, career paths, and other relevant factors.

Algorithm development: The research compares different classification algorithms and modify existing ones to better suit the task of career counseling. These algorithms are designed to handle specific challenges or considerations unique to career guidance, such as the dynamic nature of the job market or individual preferences.

2 Literature Review

In most countries, secondary schools and colleges require students to choose a topic or stream and choosing a subject or stream is important since it is the deciding factor in how far their career will go. Most of the time, a professional choice of subject or stream cannot be reversed. A poor choice of courses taken because of parental pressure, a lack of knowledge, etc. might hinder achievements in the chosen field and can also lead to failure in that field. Choosing a subject or stream based on data and experience from successful

researchers in that field and data from students about their interests, families, prior education, and other factors might increase success in that career field. Looking for a solution is developing a web-based application for high school students to address this issue. This web-based application asks a series of questions about students from several angles, revealing their interests, aptitudes, and perspectives. This application would forecast a few career alternatives based on the outcomes of the assessments. Furthermore, based on the results of the tests, the application will also inform the student's parents about their mentality. This system is developed through full-stack development. With took the help of the Django Python web application framework, SQL Lite as a database is used for backend development, whereas front-end development is done using web technologies like HTML, CSS, JavaScript, and Bootstrap. To test the website, four students were brought along. They each completed the tests independently with various inputs, and the system produced a report that was displayed to them. But a big limitation is that the students and the parents never got to know about the other options. Sometimes we make interested and dream to become something based on a particular topic is decided by many other factors, for example: looking at Neil Armstrong, one student can say I want to be an astronaut in the future but that might not be the real interest of the student and can lead to misguidance produced by the system. Therefore, if a student knows about all the possible streams and has some knowledge about the stream will help them choose wisely [3].

Guidance and counselling are the most influential parts of our lives, which we cannot ignore. And in education, we need proper guidance and counselling so that we get on the right path to reach our goals and learn more about our skills. And if we do not get proper guidance and counselling at the right time, we may also end up on the wrong path. So, here are some benefits for students to do counselling at school so that they can acquire so many things such as learning new techniques to solve problems and the ability to solve the problems in different ways. The most important thing is that they can also learn to facilitate new career options according to their skills. And they become accustomed to interrogating others. Schools and other institutions also need to hire or acquire some counsellors so that they can teach them the time management system and show them how to enjoy the freedom of their school counselling and take advantage of that according to their needs. So, in conclusion, if there is no proper guidance and counselling, the students might take the wrong path and the future is at risk. And the most significant thing is that guidance and counselling should be done at an appropriate age and time otherwise it is harmful to the next generations [4].

Career guidance is very critical because nowadays students don't know which path is correct for them to choose. These are very serious issues for them and they don't realize these issues right now. So, we need to take care of their careers and future and we are here with the proper solution it was with help of a web-based career guidance system and it gives the proper result and the accuracy was very good and it gives the proper result. While the manual career counselling is very good but sometimes the teachers were busy or they need to attend to some other students as well or their priorities will be changed so the students were not getting proper counselling, we can't let their future down so we decided to take it to the online platform and the result was commendable [5].

In today's world, students face tough competition as day-by-day students need to be very competitive. In this technical world, the students need to be very ruthless and they also need to be very clear about their career the students need to be very clear and also ploy or we can say that they need to be very organized about their career path from initial to final path. Because the competition is multiplying day by day. And if they were unable to plan their career, then their whole future is about to null. And they also need to understand that

their path must be right and that they are following the correct path. So, we are here with a great idea and its career counselling which is a great opportunity for the students in which it helps the students to develop their career path and in which we develop data administration and data handling using machine learning and these help the students to develop the best career path and the results also were good [6].

The increasing number of course options in education nowadays enables students to find the study programs of their choice. But many students end up with confusion about what to choose. The proposed Students Career Guidance software helps students to choose their career path using data mining. Many students decide on a course because of force by parents, or relatives but after they select a course that is not interesting to them, they face a lot of problems and as a result, they don't succeed in it or do it with interest. So, this software creates a system that recommends a course based on some basic details about the students like personal information, academic details, hobbies, co-curricular activities, extracurricular activities, and other activities interests, background, etc. This software uses students' activities and academic itinerary interests as inputs to recommend an appropriate study field. The inputs into the system were entered in the form of questionnaires which were later analysed and computed. As a result, students are guided to select a program that is good for them. The scope of the system can be enlarged where the system should first tell the student about all the available fields which will help them to fill in information more precisely which will increase accuracy [7].

Information Technology is a universally accepted educational instrument. It is designed to increase the efficiency and effectiveness of the educational system all around the world. Computers are widely used for instruction as well as learning in the school system. However, in addition to the utilization of Information Technology for teaching and learning, computers can also be used in career counselling. Educational counsellors can investigate the possibility of utilizing the school computer in their work to improve counsellor effectiveness and efficiency. To come up with the solution, A computer-assisted counselling package was proposed in which students are selected based on tests and helps them placing in different streams and fields. The model diagnosed students on different criteria, and also allows parents to keep informed of the student's progress and give a review to an authorized faculty member. The system also affords the counsellor to check the progress of the teachers in their respective classes and also gives Statistical Data Analysis. As a result, it provides the counsellor with a tool that can significantly cut down on time and effort spent. But this model is limited and dependent on the test to evaluate the student, choosing a field depends on many factors, and acknowledging students about the fields and streams before and then taking the test can increase efficiency [8].

Artificial intelligence is a fast-growing technology and in career guidance, it helps most students. Artificial Intelligence helps them most to get an appropriate future or we can say that it helps to build their path very easily. Although career guidance and counselling have been available since 1970, few people were aware of it until the 2000s. But after the internet was introduced those who don't know about this term career guidance and counselling are aware of them and learn so many things from them. Some of them took this as their career path and did their best. So, for this comes a method that is C3-IoC that provides a novel body of knowledge of technical and non-technical skills relevant to IT careers. And in this way, one can gain a better understanding. In conclusion, an AI-based solution called C3-IOC allows many students to get jobs and achieve their careers [9].

The evolution of job opportunities in the present industrial era has changed the need for career advice this change can be visible in each field where a student wants to make a career or in a stream, they see a safe future for themselves. So, with the use of technology

Expert System (ES) is developed to provide career guidance and course selection for a student. To make this application the Google Scholar and Science Direct research databases were used to obtain the relevant articles and data to make things easy. The expert system is specially programmed that derive their expertise from the expert of each domain or a specific field. The expert system is designed for asking questions and providing a suitable explanation to the student. These types of systems are better than conventional programs due to their capability to handle faulty reasoning and insufficient data. It is one of the most used AI techniques which is applied for minor problems and it also helps understand more how human being process a piece of information depending on their previous experience [10].

Career counselling and guidance is the most pivotal role played in students' and youth's life. This is because it gives them an appropriate path and a better future opportunity for them. So, guidance is most critical because if students and youth don't get the proper guidance and counselling then the student and youth's future is a threat to society too. Hence, now as the internet introduced, this problem is also solved and many of them get proper outputs from them. So, now students and youths also don't need counselling in a physical sense; they have their solution on the internet and they have numerous solutions to their single problem. In conclusion, we can say that there is an advantage to the Internet but it has a disadvantage too because some of the students don't know what to search for on the Internet so physically, they have the necessary attention and guidance and were be more comfortable with them than the internet [11].

The study is based on e-counselling services in today's digital era, in the study, researchers did a survey on 348 male and teachers of Jordan at Rusaifa education directorate in 2021. The study reaches some statistics taken from teachers' perspectives. Nowadays teaching is witnessing a notable shift due to corona pandemic which leads to a major conversion of information and communication technology in the education system to handle this change and keep track of it researchers conduct this survey, and after this survey, researchers think to deal with this shift of technology and help students with the use of modern technological tools. Regarding these changes we live, the need for psychological counselling, and counselling should keep pace with today's digital era and help students understand themselves. As a result, the requirement for school counsellors must use digital tools to help them complete daily tasks and other services, also improving clarity related to choosing streams and the social and emotional needs of students. To maintain these findings a strong commitment is required to improve schoolwork, and teachers must be aware of to guide students according to counsellor-provided services [12].

Web-based and app based career guidance is an important aspect of today's education system [13–16]. In the existing system, we do things manually with human counsellors but it has a few problems when it comes to the scalability of career guidance and also for student comfort. Here scalability means the number of counsellors available in schools, universities, and other educational institutions. Generally, students face problems reaching counsellors so to help the students researchers created a counselling system to provide services to students of Nigeria studying in various institutions. The system was developed with the help of a survey taken of students and counsellors. The system is designed with the help of scripting languages like PHP, HTML, JavaScript, and CSS, and MySQL is used for Data derived from this survey. The implementation of the system was trained and tested with 50 students out of which 76% were satisfied by the system [17].

As we all know that in the past students picked their career according to their parent's choice but now the time has completely changed because these generations can't choose their career according to their parent's choice, they want to choose their career on their

own. With the advent of revolutionary technologies like Machine Learning, AI, Blockchain and many more career options in it enables youngsters to choose among wide variety of career options [18–23]. So, this rift between students and parents we need to take this as a priority and we need to solve their problems with the help of career counselling with the help of AI because now parents too have knowledge of the internet and they can easily browse what is correct for their children. As a result of AI and chatbots in career counselling, parents take it very seriously and become involved in their children's careers as well as knowing what their children are interested in. With the chatbot in artificial intelligence, parents also have relief that their child doesn't choose the wrong career in the future. Parents can now make well-informed decisions about their child's career, as they can access information regarding various career paths. AI and chatbots can provide up-to-date data to parents. They can also help guide children in making the right career choices [24].

This Covid-19 pandemic changed everyone's life and changed the way we look at technology [25, 26]. Now technology reaches its peak because Covid-19 shows that you can't depend on each other and you need to do your work So, ICT (Information and Communication Technology) and AI (Artificial Intelligence) both are at the top of their game and trend. In career guidance and counselling the role of AI is very significant because now there is no one around you always discussing your career. This is because they also have their priorities so they were busy. As a result of these problems, AI now provides you with career counselling, career information, career learning, or matching you to a suitable occupation. In recent years, AI has been deployed in fields with similar social challenges. Given the current lack of research specific to career counselling, experience from these analogous settings can shed some light on career-related applications. These fields are therapeutic counselling, library and information services, education, and employee selection. AI can be used to provide automated, personalized advice by analysing data from career assessments, labour market trends, and job postings. AI can also be used to help with career exploration, providing detailed and interactive visualizations of different career paths. Finally, AI can be used to assist in job searches by providing tailored job recommendations [27].

The focal point of South Africa's higher education in the twenty-first century is the transformation of subjects. This is seen as necessary as it addresses social concerns such as socio-economic, educational, and health inconsistencies by addressing issues of access, equity, and diversity. The transformation of higher education in South Africa also includes the restructuring of courses, the introduction of new courses, and the redesign of curricula. This is done to ensure that students are equipped with the skills and knowledge necessary to tackle the challenges of the modern world. In considering the education system, South African student's higher education is consistently dropping. Undergraduate drop-outs are increasing and it is about 52% of those who are getting career counselling and guidance proper education rates are 34% and the rest don't want to study anymore because they felt that they have enough education. This indicates a need for systemic reform in the education system. The government must prioritize providing career counselling and guidance to students to ensure that they are adequately prepared to meet the demands of the contemporary world. Additionally, the government must also create an education system that encourages and incentivizes students to stay in school. This will help students to make informed decisions about their future. It will also enable them to gain the skills necessary to succeed in the job market. Finally, it will empower them to make a positive contribution to society [28].

Everyone uses electronic gadgets in today's world, and the growing impact of the smartphone increases thus the development of a mobile application for educational Guidance and Counselling will help students for better and more convenient guidance. The college

counselling Recommendation System helps students who have been checking the college availability for their cut-off marks and location, using Android technology [29].

As we all know, career-related confusion among students is nowadays common because at the age of 18 is common because students don't know what is best for them, they thought that whatever their friends do is right for them and they make it that after some age they realized that it is the wrong path for them. A recent survey of 115,000 students of 33 different ages chose an incorrect career or we can say that they choose the wrong path which is Approx. 50% and we need to fix this because now this percentage is increasing. This is concerning because making the wrong career choice can have long-term negative impacts on a person's life. To help address this issue, it is important to provide students with better guidance and access to resources that can help them make informed decisions about their future. Schools should offer career counselling services and resources to support students in making informed decisions. Additionally, employers should also develop more resources to help students make informed decisions. This can be done by providing mentorship and internship programs and offering more scholarships and grants. Government agencies should also provide more job training and education programs to equip students with the necessary skills for their chosen career paths. Finally, society should continue to support students in pursuing their goals by creating more opportunities for them [30].

Today as we all know; career counselling is also a very big challenge and right now students also facing very big problems regarding career counselling because they also don't know whom to trust or not because it's a very big decision regarding their career. And the students are facing problems like economic, globalization, and technological advancement. And the most important thing is that nowadays, competition in studies is now a very important issue and the students were scared. So, we are with a solution that students will no longer have to take strain because we are with machine learning algorithms that helps them to take the proper decision and this algorithm gives them proper accuracy so that they can take the decision wisely. The algorithms are accurate, quick, and easy to use, and can be tailored to an individual's needs. This will allow students to make informed decisions in a fraction of the time, taking the stress out of the decision-making process. It will also help to level the playing field, ensuring that all students have access to the same level of information. This will enable students to make decisions that are in line with their strengths, interests, and goals. It will also help reduce the gap between those with access to resources and those without. This will give students the ability to make decisions that are tailored to their individual needs, ensuring they have the best chance of success. It will also help to create a more equitable education system as all students can access the same resources [31].

As we all know, nowadays the education system is very complex and very vast. The students are the most significant in the education system and major contributors to the education system so, so we need to focus on their careers and care for them a career that they don't drag too wrong. To ensure that students make the right career choices and do not become overwhelmed by the vastness of the system, it is important to provide them with access to resources and guidance, such as career counselling and support services. This will enable them to make informed decisions about their future and ensure that they pursue the career path that best suits their interests and skills. Students should also be encouraged to explore different career options and research the industries they are interested in. Additionally, they should be allowed to gain experience in their chosen field through internships and other work-related activities. This will help them to have a better understanding of the job market and help them to make a more informed decision. They should also reach out to professionals in the field to gain insight into their

experience and advice. Finally, they should not be afraid to take risks and try something new [32].

As we all know, career-related confusion among students is nowadays common because, at the age of 18, it is common because students don't know what is best for them, they thought that whatever their friends do right for them and they make after some age they realized that it is the wrong path for them. A recent survey of 115,000 students from 33 different ages chose an incorrect career or we can say that they choose the wrong path which is Approx. 50% and we need to fix this because now this percentage is increasing. This is concerning because making the wrong career choice can have long-term negative impacts on a person's life. To help address this issue, it is important to provide students with better guidance and access to resources that can help them make informed decisions about their future. Schools should offer career counselling services and resources to support students in making informed decisions. Additionally, employers should also develop more resources to help students make informed decisions. This can be done by providing mentorship and internship programs and offering more scholarships and grants. Government agencies should also provide more job training and education programs to equip students with the necessary skills for their chosen career paths. Finally, society should continue to support students in pursuing their goals by creating more opportunities for them [33].

In today's fast-paced world, students need to recognize the value of career counselling in building a successful future. Seeking guidance from experienced professionals who understand the job market and the skills required for success is crucial. The right career guidance can empower students to make informed decisions about their career path, avoiding costly mistakes and maximizing their potential. Having a mentor or career coach to provide advice and resources is invaluable. They can offer strategies for job searching, interviewing, networking, and more. They can also provide insight into the current job market and help students identify the skills and qualifications that employers are seeking. In addition, mentors and career coaches can provide emotional support, motivation, and guidance throughout the job search process. Identifying strengths and weaknesses and offering advice on skill development is another valuable aspect of career counselling. Mentors and coaches can help students hone their skills and provide guidance on how to navigate the job market, including tailoring resumes to specific job postings and creating effective networking strategies. They can also provide students with the tools and resources needed for success, such as interviewing techniques and creating a professional portfolio. In summary, career counselling is essential for students to make informed decisions about their future and achieve success. Having experienced mentors or career coaches to provide guidance, support, and resources can be instrumental in helping students build a successful career [34].

The use of machine learning models to predict careers has proved to be a promising approach. Previously also so much work had been on career prediction, using different techniques which is mentioned in Table 1. There is also much work stating the need for career counselling at different stages of academics. The right career guidance can change the story of many students. Many students pursue guidance from some individual or randomly from the internet, which leads to different opinions and confusion. Before choosing a career option it is really necessary to know different aspects of the option such as its market value, future scope, expected earning, the interest of the student in that field, etc. As per our research, many digital tools in the market give students a career option to pursue based on the questionnaire filled, but telling them about possible career options systematically will help the user to think and system to produce more accurate results.

Table 1 Analytical literature summary

S. No	Source information	Work done	Techniques used	Findings	Future scope
1	Empowering high school students with web based career counselling [1]	Created a website for career counselling	Django Python web application framework, SQL Lite as database, HTML, CSS, JavaScript, and bootstrap	The system gave the guidance and suggested the market value of that field	To provide students of developing countries the flexibility to pick a career option
2	Exploring career guidance expert systems for students: ODL implications [8]	An Expert system was made with the help of the experts in the field to reflect that expertise in the system and to benefit students with a single platform	The methods used are rule-based, case-based, and fuzzy logic-based expert systems used for the career guidance of the students	The authors observed that the rule-based career guidance systems are most popular due to the ease with which these can be designed	The aim of guidance and counselling is fulfilling the student's physiological needs and self-evaluation. And to be helpful for various decision-makers in institutions
3	Web-enabled scientific career counselling system [3]	Created system that provides a suitable recommendation for candidates to choose appropriate Job Roles	To develop this mode JavaScript, HTML, MySQL, CSS, NODE JS, EXPRESS, Bootstrap are used	It predicts suitable department in accordance to the performance of the candidate	The scope of the Project can be extended to encompass additional areas of prediction, such as forecasting the career trajectory of candidates after completing their 12th or 10th grade education

Table 1 (continued)

S. No	Source information	Work done	Techniques used	Findings	Future scope
4	Smart career advisory system with recommendations [4]	Created Student Career Guidance and Recommendation System	Machine learning techniques such as Support Vector Machines (SVM), Random Forests, Decision Trees, One Hot Encoding, XG Boost, Deep Neural Networks, and Time Series Analysis are employed in the Project	The system gives guidance and recommendation for career	The Project aims to develop a robust web application that collects information through thorough evaluation and examination. This includes assessing various aspects such as analytical skills, memory capabilities, technical proficiency, logical reasoning, hobbies, interests (both technical and non-technical), as well as the student's performance history from childhood and skill-based tests. The data gathered will be utilized to enhance the accuracy and effectiveness of the application
5	Machine learning-driven career guidance system for skill assessment and network visualization [7]	Created an AI-Based solution for students to explore a career path in IT	Developed a C3-IoC method which presents a novel similarity metric method	Give users the best possible options by Creating job repositories and giving users skills needed in the field they want to work	In today's emerging market situation, this model will provide skills according to market or job changes
6	Web-enabled career guidance system for students in Nigeria's pre-tertiary institutions [11]	"A web Based career guidance system with human counselors in charge."	Used PHP, MySQL, HTML, JavaScript, and CSS to develop an application	The System was Trained and Tested with 50 students and 76% of accuracy was found	The career guidance system is an accessible tool for real-life counsellors and students and people in any profession
7	Artificial intelligence-based online career counselling system: a novel approach [12]	Created an intelligence chatbot that predicts the performance of students	Developed a chatbot using natural language processing to communicate with the user and a fuzzy expert system that helps in the prediction	This system provides accurate options to the user using an intelligent engine	The career counsellor system with AI using a fuzzy expert system will play a major role in the upcoming world

Table 1 (continued)

S. No	Source information	Work done	Techniques used	Findings	Future scope
8	Prospects and perils of AI in career development practice [13]	Automated Therapeutic counselling is created and it is used to help user form their personal issues	Used AI technology for students to enhance their capacity to gather accurate information for clients the system provides risk and benefits to clients according to ongoing circumstances	The system has the ability to act autonomously when providing client services which helps the client to see the whole system more as a real counsellor rather than an AI tool	The AI system provided is not developed as a project it is made to use in service design and AI is used for automated interaction rather than as a tool
9	Assessing Artificial Intelligence in Career Guidance and Counselling: A Comprehensive Appraisal [17]	Provided a solution to students using expert system to help them in their career selecting experience	Expert System is used that offers solution to difficult decision-making problems depending on the information gained	81% accuracy with ANN Classification	Aims to use this career counselling platform centered on AI techniques to safeguard future of students
10	Smart Learning Environment: AI-Based Monitoring and Preventive Measures for Student Performance [18]	Made a prediction model to make it easy for instructors to help students with an unsatisfactory career choice	The prediction model is made with distinct machine learning algorithms like a decision tree, KNN, ANN, and Naive Bayes	The accuracy of all prediction models is: KNN = 82.8 Decision Tree = 86.1 ANN = 84.8 Naive Bayes = 84.1	The aim is to extend the notions and apply the model to more courses to increase its efficiency of the model

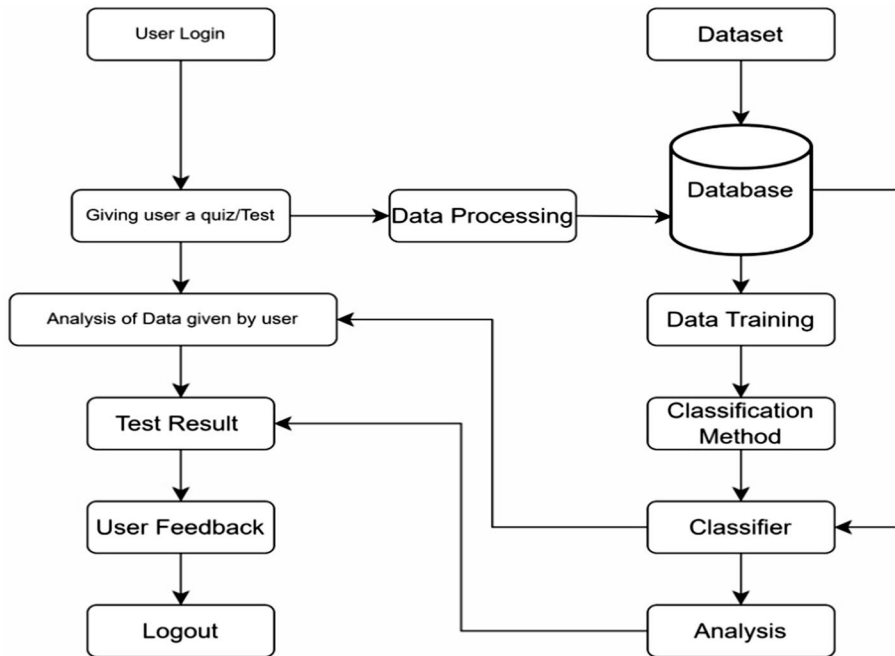
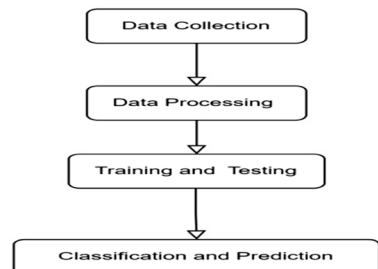


Fig. 1 Proposed methodology

Fig. 2 Process flow diagram



3 Proposed Methodology

Figures 1, 2 depicted the proposed methodology and process flow diagram.

- Data collection*: The first step is to gather data about the user, including their interests, skills, experience, education, and career goals. This will be done through questionnaires.
- Data Pre-processing*: Data must be in organized form, therefore cleaning of all the data, replacing them with appropriate or approximate data and also removing null data and replacing them with some fixed alternate values are some of the basic steps in preprocessing of data.
- Division of Dataset*: From the huge data set collected 80 percent of the data is utilized for training and 20 percent of the data is reserved for testing. If maximum number of

predictions is right then model will have a good accuracy percentage and is reliable to continue with otherwise better to change the model.

The column names of dataset taken into account had slightly changed to get proper understanding of the data. The dataset having 20 features along with 6902 records with preprocessing [35].

- d. Model Building using Decision Tree, Random Forest, Support Vector Machine (SVM), Naïve Bayes, KNN

- i. Training with Decision Tree Classifier

1. Load the Pandas library and the dataset using Pandas.
2. Define the features and the target variables.
3. Inspect the features and target variables to understand the data.
4. Split the dataset into training and testing sets using Sklearn's `train_test_split` function.
5. Import the `DecisionTreeClassifier` function from the Sklearn tree module and create a Decision Tree model.
6. Train the Decision Tree model on the training data.
7. Evaluate the performance of the Decision Tree model using appropriate evaluation metrics such as accuracy, precision, recall, or F1-score.
8. Optionally, perform hyperparameter tuning to optimize the performance of the Decision Tree model.
9. Visualize the Decision Tree using Sklearn's `plot_tree` function to understand the learned decision rules and feature importance.
10. Interpret the results and make conclusions based on the performance and interpretability of the Decision Tree model.

- ii. Training with Random Forest

1. Load the Pandas library and the dataset using Pandas.
2. Define the features and the target variables.
3. Inspect the features and target variables to understand the data.
4. Split the dataset into training and testing sets using Sklearn's `train_test_split` function.
5. Import the `RandomForestClassifier` function from the Sklearn ensemble module and create a Random Forest model.
6. Train the Random Forest model on the training data.
7. Evaluate the performance of the Random Forest model using appropriate evaluation metrics such as accuracy, precision, recall, or F1-score.
8. Optionally, perform hyperparameter tuning to optimize the performance of the Random Forest model.
9. Visualize the feature importance using the `feature_importances_` attribute of the trained Random Forest model to understand the relative importance of different features.
10. Interpret the results and make conclusions based on the performance and interpretability of the Random Forest model.

- iii. Training with Support Vector Machine (SVM)

1. Load the Pandas library and the dataset using Pandas.
2. Define the features and the target variables.
3. Inspect the features and target variables to understand the data.
4. Split the dataset into training and testing sets using Sklearn's `train_test_split` function.
5. Import the SVC (Support Vector Classifier) function from the `Sklearnsvm` module and create an SVM model.
6. Train the SVM model on the training data.
7. Evaluate the performance of the SVM model using appropriate evaluation metrics such as accuracy, precision, recall, or F1-score.
8. Optionally, perform hyperparameter tuning to optimize the performance of the SVM model, such as tuning the kernel type (linear, polynomial, radial basis function (RBF), etc.), regularization parameter (C), and gamma.
9. Interpret the results and make conclusions based on the performance and interpretability of the SVM model.
10. Optionally, visualize the decision boundary of the SVM model to understand how it separates the classes in the feature space, especially for binary classification problems.
11. If needed, you can also perform feature scaling or other data preprocessing steps to improve the performance of the SVM model.

iv. Training with Naïve Bayes

1. Load the Pandas library and the dataset using Pandas.
2. Define the features and the target variables.
3. Inspect the features and target variables to understand the data.
4. Split the dataset into training and testing sets using Sklearn's `train_test_split` function.
5. Import the NaiveBayes classifier (such as GaussianNB for continuous features or MultinomialNB for discrete features) from the `Sklearnnaive_bayes` module and create a Naive Bayes model.
6. Train the Naive Bayes model on the training data.
7. Evaluate the performance of the Naive Bayes model using appropriate evaluation metrics such as accuracy, precision, recall, or F1-score.
8. Optionally, perform any necessary data preprocessing steps, such as handling missing values, transforming features, or handling categorical variables, to prepare the data for the Naive Bayes model.
9. Interpret the results and make conclusions based on the performance and interpretability of the Naive Bayes model.
10. Optionally, perform feature selection or feature extraction techniques to improve the performance of the Naive Bayes model.
11. If needed, you can also perform hyperparameter tuning to optimize the performance of the Naive Bayes model, such as tuning the smoothing parameter for MultinomialNB or adjusting priors for GaussianNB.
12. Optionally, perform cross-validation to assess the generalization performance of the Naive Bayes model on unseen data.

v. Training with KNN

1. Load the Pandas library and the dataset using Pandas.
2. Define the features and the target variables.
3. Inspect the features and target variables to understand the data.
4. Split the dataset into training and testing sets using Sklearn's `train_test_split` function.
5. Import the `KNeighborsClassifier` function from the `Sklearnneighbors` module and create a KNN model.
6. Train the KNN model on the training data.
7. Evaluate the performance of the KNN model using appropriate evaluation metrics such as accuracy, precision, recall, or F1-score.
8. Optionally, perform hyperparameter tuning to optimize the performance of the KNN model, such as tuning the number of neighbors (K), the distance metric, or other parameters.
9. Optionally, perform any necessary data preprocessing steps, such as handling missing values, scaling features, or handling categorical variables, to prepare the data for the KNN model.
10. Interpret the results and make conclusions based on the performance and interpretability of the KNN model.
11. Optionally, perform feature selection or feature extraction techniques to improve the performance of the KNN model.
12. If needed, you can also perform cross-validation to assess the generalization performance of the KNN model on unseen data.
13. Optionally, visualize the decision boundaries or other relevant information to understand how the KNN model makes predictions in the feature space.
14. Interpret the results and make conclusions based on the performance and interpretability of the KNN model.

e. Analysis of Accuracy/Evaluation

Accuracy measures the number of correct predictions made by a machine learning model in relation to the total number of predictions made. We calculate it by dividing the number of correct predictions by the total number of predictions.

Accuracy = Number of correct predictions / Total number of predictions

Mathematically, it can be expressed as:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

where:

1. TP (True Positive) is the number of instances that are correctly predicted as positive.
2. TN (True Negative) is the number of instances that are correctly predicted as negative.
3. FP (False Positive) is the number of instances that are incorrectly predicted as positive (i.e., instances that are actually negative but predicted as positive).
4. FN (False Negative) is the number of instances that are incorrectly predicted as negative (i.e., instances that are actually positive but predicted as negative).

4 Design and Implementation

First, we will import all the required Libraries and the dataset. Then Data Processing will be done, in which checking of missing values, distinct values, data balance for classification will take place. After that following will take place. The Fig. 3 depicted the correlation between the numerical features and Figs. 4, 5, 6, 7, 8 and 9 depicted the categorical data visualization.

a. Exploratory Data Analysis

- Correlation Between Numerical Features: No highly correlated numerical pair found.
- Categorical Data Visualization

b. Building Machine Learning Models

The different machine learning build in this works the decision tree classifier, random forest classifier, SVM, Naive Bayes and KNN is depicted in Figs. 10, 11 12, 13 14, 15 16, 17 and 18 respectively. The output of the implement model the decision tree classifier, random forest classifier, SVM, Naive Bayes and KNN is depicted in Figs. 11, 13, 15, 17 and 19 respectively.

- Decision Tree Classifier:

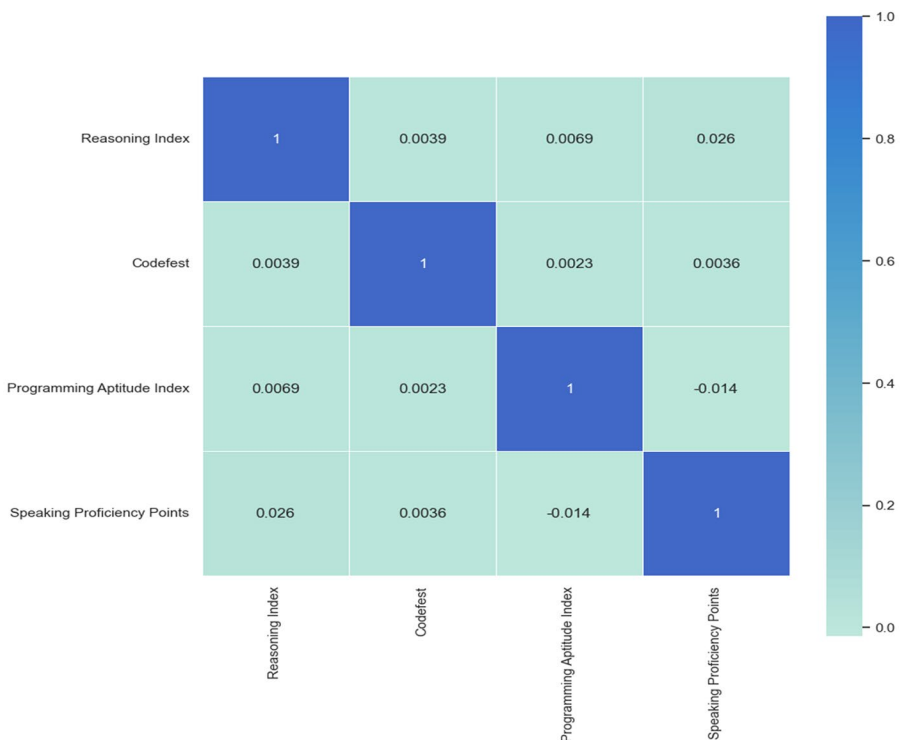


Fig. 3 Representation of correlation between numerical features

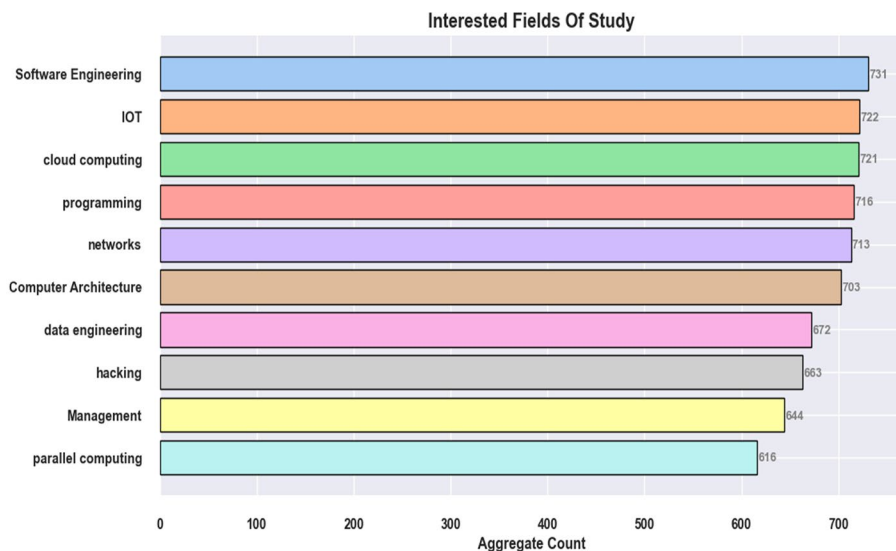


Fig. 4 Represents interested fields of study

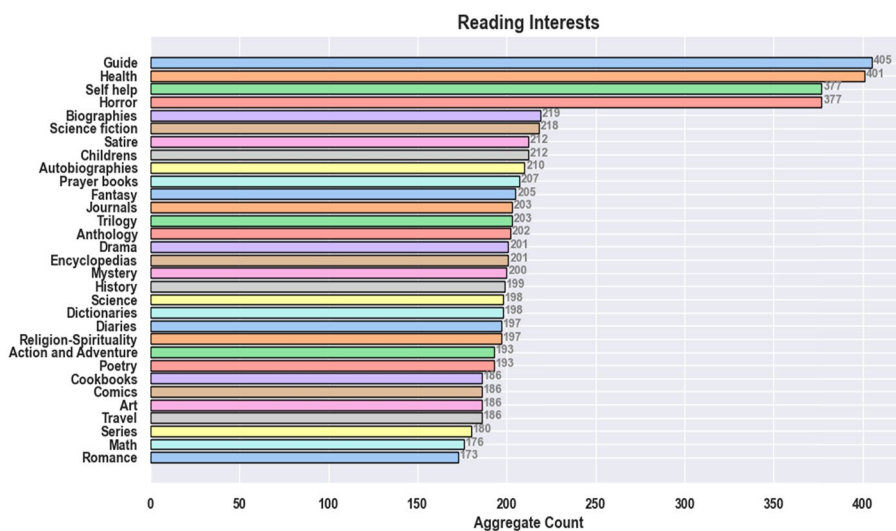


Fig. 5 Represents reading interests in different fields

After running the code, we found that the accuracy was coming out to be 66.33%. We also gave some manual inputs to the system to check the manual accuracy and the predicted output.

Manual input and output: In Table 2, out of 7 outputs, 4 inputs match the actual output, thus giving the Accuracy of 57.14%.

- Random Forest Classifier:

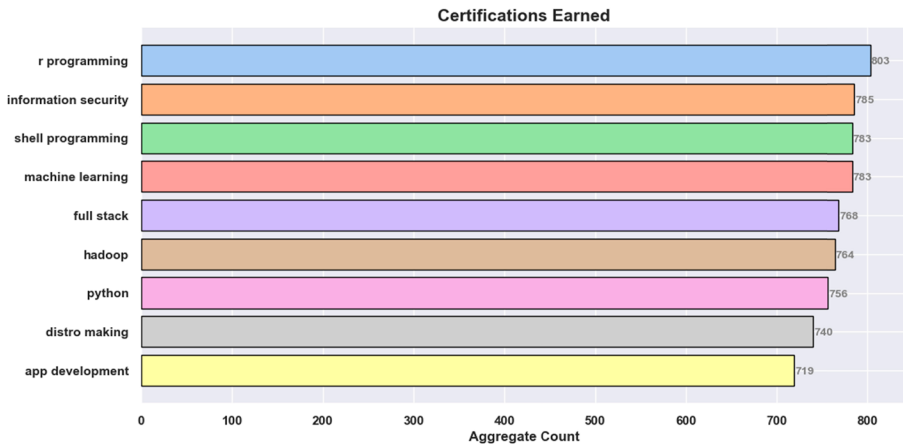


Fig. 6 Represents certifications earned in different fields

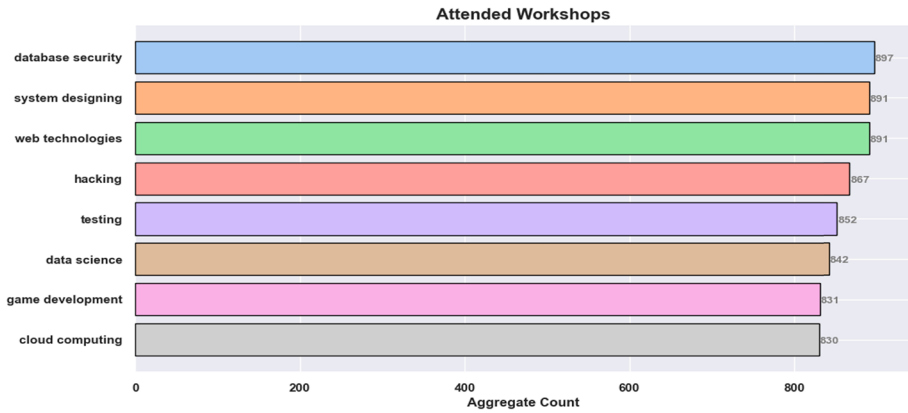


Fig. 7 Represents attended workshops in different fields

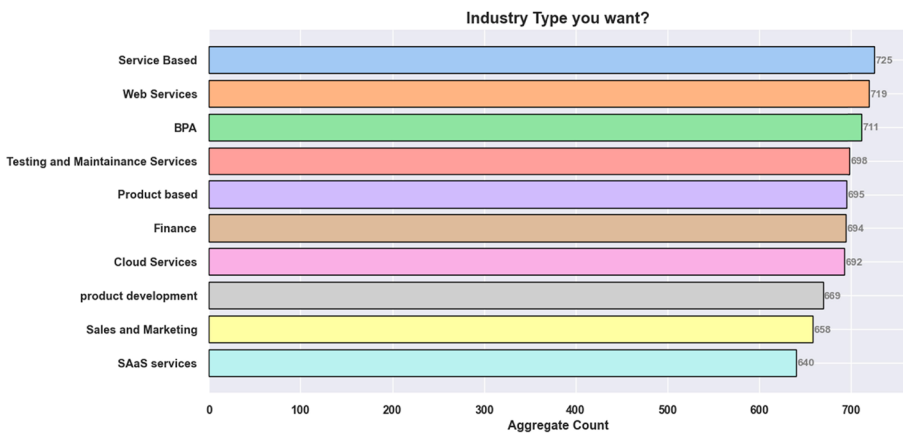


Fig. 8 Represents industry type in which one wants to work

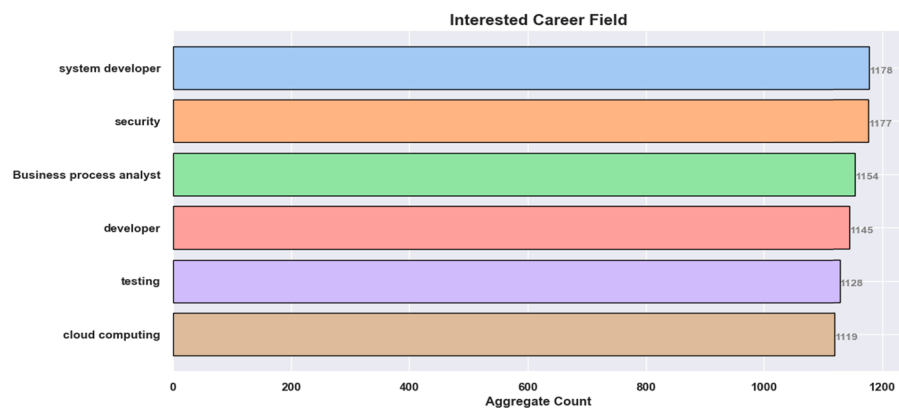


Fig. 9 Represents interested career fields

- Decision Tree Classifier:

```
In [54]: from sklearn.metrics import confusion_matrix, accuracy_score
         from sklearn.model_selection import train_test_split
         from sklearn.tree import DecisionTreeClassifier

         # Create and fit the classifier
         classifier = DecisionTreeClassifier(random_state=1)
         classifier.fit(x_train, y_train)

         # Make predictions
         y_pred = classifier.predict(x_test)

         # Compute accuracy
         accuracy = accuracy_score(y_test, y_pred)

         # Print the results
         print("\nAccuracy:", accuracy * 100, "%")
```

Fig. 10 Implementation of decision tree classifier

```
In [60]: import numpy as np

         userdata = [['2', '4', '6', '8', '10', '12', '14', '16', '18', '20', '22', '24', '26', '28',
                     '30', '32', '34', '36', '38', '40', '42']]
         ynewclass = dtree.predict(userdata)
         ynew = dtree.predict_proba(userdata)
         print(ynewclass)
         print("Probabilities of all classes: ", ynew)
         print("Probability of Predicted class : ", np.max(ynew))

         ['Systems Security Administrator']
```

Fig. 11 System output based on user input for decision tree classifier

After running the code, we found that the accuracy was coming out to be 84.17%. We also gave some manual inputs to the system to check the manual accuracy and the predicted output.

- Random Forest Classifier:

```
In [56]: from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

# Create and fit the Random Forest classifier
classifier = RandomForestClassifier(random_state=1)
classifier.fit(x_train, y_train)

# Make predictions
y_pred = classifier.predict(x_test)

# Compute accuracy
accuracy = accuracy_score(y_test, y_pred)

# Print the results
print("Random Forest Accuracy:", accuracy * 100, "%")
```

Fig. 12 Implementation of random forest classifier

Manual input and output:

```
In [41]: userdata = [['7', '6', '6', '8', '3', '5', '4', '4', '7', '3', '3', '6', '8',
                    '7', '5', '7', '4', '5', '6', '8', '8']]
ynewclass = rf.predict(userdata)
ynew = rf.predict_proba(userdata)
print(ynewclass)
print("Probabilities of all classes: ", ynew)
print("Probability of Predicted class :", np.max(ynew))

['Web Developer']
```

Fig. 13 System output based on user input for random forest classifier

```
In [55]: from sklearn.svm import SVC
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

# Create and fit the SVM classifier
classifier = SVC(random_state=1)
classifier.fit(x_train, y_train)

# Make predictions
y_pred = classifier.predict(x_test)

# Compute accuracy
accuracy = accuracy_score(y_test, y_pred)

# Print the results
print("SVM Accuracy:", accuracy * 100, "%")
```

Fig. 14 Implementation of SVM

Manual input and output: In Table 3, out of 7 outputs, 6 inputs match the actual output, thus giving the Accuracy of 85.71%.

- SVM:

Manual input and output:

```
In [48]: userdata = [['21','20','19','18','17','16','15','14','13','12','11','10','9','8',
'7','6','5','4','3','2','1'
]]
ynewclass = svm.predict(userdata)
ynew = svm.decision_function(userdata)
print(ynewclass)
print("Probabilities of all classes: ", ynew)
print("Probability of Predicted class : ", np.max(ynew))

['Software Engineer']
```

Fig. 15 System output based on user input For SVM

- Naïve Bayes:

```
In [57]: from sklearn.naive_bayes import GaussianNB
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

# Create and fit the Naive Bayes classifier
classifier = GaussianNB()
classifier.fit(x_train, y_train)

# Make predictions
y_pred = classifier.predict(x_test)

# Compute accuracy
accuracy = accuracy_score(y_test, y_pred)

# Print the results
print("Naive Bayes Accuracy:", accuracy * 100, "%")
```

Fig. 16 Implementation of naïve bayes

Manual input and output:

```
# Make predictions on new data
userdata = [[6, 7, 6, 7, 6, 7, 6, 7, 6, 7, 6, 7, 6, 7, 6, 7, 6, 7, 6]]
ynewclass = gnb.predict(userdata)
ynew = gnb.predict_proba(userdata)
print("Predicted class: ", ynewclass)
print("Probabilities of all classes: ", ynew)
print("Probability of Predicted class: ", ynew.max())

Predicted class: ['Web Developer']
```

Fig. 17 System output based on user input for naïve bayes

- KNN:

```
In [58]: from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

# Create and fit the K-nearest neighbors classifier
classifier = KNeighborsClassifier()
classifier.fit(x_train, y_train)

# Make predictions
y_pred = classifier.predict(x_test)

# Compute accuracy
accuracy = accuracy_score(y_test, y_pred)

# Print the results
print("K-nearest neighbors Accuracy:", accuracy * 100, "%")
```

Fig. 18 Implementation of KNN

```
# Define the user data for prediction
userdata = np.array([
    [21, 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1]
])

# Predict the class labels for the user data
ynewclass = knn.predict(userdata)

# Print the predicted class labels
print("Predicted class labels: ", ynewclass)

Predicted class labels: ['Database Developer']
```

Fig. 19 System output based on user input for KNN

After running the code, we found that the accuracy was coming out to be 75.81%. We also gave some manual inputs to the system to check the manual accuracy and the predicted output.

Manual input and output: In Table 4, out of 7 outputs, 5 inputs match the actual output, thus giving the Accuracy of 71.42%.

- Naïve Bayes:

After running the code, we found that the accuracy was coming out to be 69.93%. We also gave some manual inputs to the system to check the manual accuracy and the predicted output.

Manual input and output: In Table 5, out of 7 outputs, 5 inputs match the actual output, thus giving the Accuracy of 71.42%.

- KNN:

After running the code, we found that the accuracy was coming out to be 70.84%. We also gave some manual inputs to the system to check the manual accuracy and the predicted output.

Manual input and output: In Table 6, out of 7 outputs, 5 inputs match the actual output, thus giving the Accuracy of 71.42%.

Table 2 Manual accuracy of decision tree classifier

Manual input given by user	Output generated by our model for given input	Actual output of the given input	Result
'7','6','6','8','3','5','4','4','7','3','3','6','8','7','5','7','4','5','6','8','8'	Web developer	Web developer	True
'1','2','3','4','5','6','7','8','9','10','11','12','13','14','15','16','17','18','1','20','21'	Software developer	Software developer	True
'10','9','8','7','6','5','4','3','2','1','10','9','8','7','6','5','4','3','2','1','10'	CRM technical developer	CRM technical developer	True
'2','4','6','8','10','12','14','16','18','20','22','24','26','28','30','32','34','36','38','40','42'	System security administrator	Software developer	False
'3','3','3','3','3','3','3','3','3','3','3','3','3','3','3','3','3','3','3','3'	Technical support	UX designer	False
'6','7','6','7','6','7','6','7','6','7','6','7','6','7','6','7','6','7','6','7'	Web developer	Web developer	True
'21','20','19','18','17','16','15','14','13','12','11','10','9','8','7','6','5','4','3','2','1'	Technical support	Database developer	False

Table 3 Manual accuracy of random forest classifier

Manual input given by user	Output generated by our model for given input	Actual output of the given input	Result
'7','6','8','3','5','4','4','7','3','3', '6','8','7','5','7','4','5','6','8','8'	Web developer	Web developer	True
'1','2','3','4','5','6','7','8','9','10','11','12','13','14','15', '16','17','18','19','20','21'	Software developer	Software developer	True
'10','9','8','7','6','5','4','3','2','1','10','9','8','7','6','5', '4','3','2','1','10'	CRM technical developer	CRM technical developer	True
'2','4','6','8','10','12','14','16','18','20','22','24','26','28', '30','32','34','36','38','40','42'	Software developer	Software developer	True
'3','3','3','3','3','3','3','3','3','3','3','3','3','3','3', '3','3','3','3','3'	UX designer	UX designer	True
'6','7','6','7','6','7','6','7','6','7','6','7','6','7', '6','7','6','7','6'	UX designer	Web developer	False
'21','20','19','18','17','16','15','14','13','12','11','10','9','8', '7','6','5','4','3','2','1'	Database developer	Database developer	True

Table 4 Manual accuracy of SVM

Manual input given by user	Output generated by our model for given input	Actual output of the given input	Result
'7','6','8','3','5','4','4','7','3', '3','6','8','7','5','7','4','5','6','8','8', '1','2','3','4','5','6','7','8','9','10','11','12','13','14','15', '16','17','18','19','20','21', '10','9','8','7','6','5','4','3','2','1','10','9','8','7','6','5', '4','3','2','1','10', '2','4','6','8','10','12','14','16','18','20','22','24','26','28', '30','32','34','36','38','40','42', '3','3','3','3','3','3','3','3','3','3','3','3','3','3','3', '3','3','3','3','3', '6','7','6','7','6','7','6','7','6','7','6','7','6','7', '6','7','6','7','6', '21','20','19','18','17','16','15','14','13','12','11','10','9','8', '7','6','5','4','3','2','1',	Web developer Software developer CRM technical developer Software developer UX designer UX designer Software engineer	Web developer Software developer CRM technical developer Software developer UX designer Web developer Database developer	True True True True True False False

Table 5 Manual accuracy of naïve Bayes

Manual input given by user	Output generated by our model for given input	Actual output of the given input	Result
'7','6','6','8','3','5','4','4','7','3','3','6','8','7','5','7','4','5','6','8','8'	Web developer	Web developer	True
'1','2','3','4','5','6','7','8','9','10','11','12','13','14','15','16','17','18','19','20','21'	Software developer	Software developer	True
'10','9','8','7','6','5','4','3','2','1','10','9','8','7','6','5','4','3','2','1','10'	CRM technical developer	CRM technical developer	True
'2','4','6','8','10','12','14','16','18','20','22','24','26','28','30','32','34','36','38','40','42'	UX designer	Software developer	False
'3','3','3','3','3','3','3','3','3','3','3','3','3','3','3','3','3','3','3','3'	UX designer	UX designer	True
'6','7','6','7','6','7','6','7','6','7','6','7','6','7','6','7','6','7','6','7'	Web developer	Web developer	True
'6','7','6','7','6'	Web developer	Database developer	False
'21','20','19','18','17','16','15','14','13','12','11','10','9','8','7','6','5','4','3','2','1'			

Table 6 Manual accuracy of KNN

Manual input given by user	Output generated by our model for given input	Actual output of the given input	Result
'7','6','6','8','3','5','4','4','7','3','3','6','8','7','5','7','4','5','6','8','8'	Web developer	Web developer	True
'1','2','3','4','5','6','7','8','9','10','11','12','13','14','15','16','17','18','19','20','21'	Database developer	Software developer	False
'10','9','8','7','6','5','4','3','2','1','10','9','8','7','6','5','4','3','2','1','10'	CRM technical developer	CRM technical developer	True
'2','4','6','8','10','12','14','16','18','20','22','24','26','28','30','32','34','36','38','40','42'	Database developer	Software developer	False
'3','3','3','3','3','3','3','3','3','3','3','3','3','3','3','3','3','3','3','3'	UX designer	UX designer	True
'6','7','6','7','6','7','6','7','6','7','6','7','6','7','6','7','6','7','6','7'	Web developer	Web developer	True
'21','20','19','18','17','16','15','14','13','12','11','10','9','8','7','6','5','4','3','2','1'	Database developer	Database developer	True



Fig. 20 Decision tree classifier accuracy graph



Fig. 21 Random forest classifier accuracy graph

5 Results and Discussion

The accuracy of machine learning model decision tree classifier, random forest classifier, SVM, Naive Bayes and KNN is depicted in Figs. 20, 21, 22, 23 and 24 respectively.

a. Decision Tree Classifier



Fig. 22 SVM classifier accuracy graph

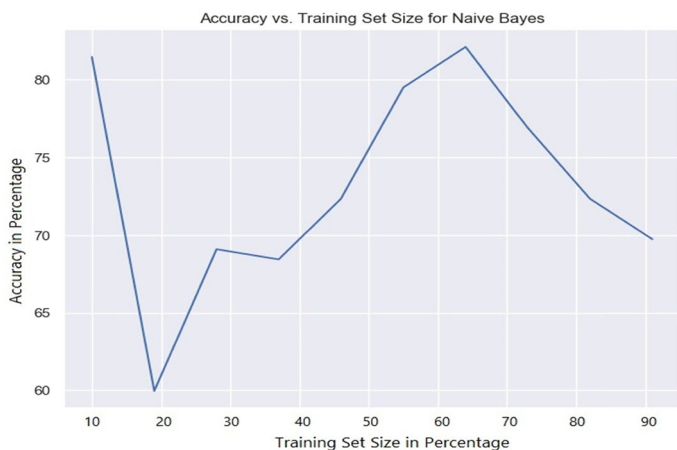


Fig. 23 Naïve bayes classifier accuracy graph

After implementing Decision Tree Classifier to predict the right career stream, we found that the classifier accuracy was coming out to be 66.33%. We also took manual inputs and got accuracy of 57.41%.

Optimizing Decision Tree Performance: Exploring Hyperparameter Tuning Techniques:



Fig. 24 KNN classifier accuracy graph

For hyperparameter tuning we have used grid search to tune the `max_depth`, `min_samples_split`, and `min_samples_leaf` hyperparameters of a decision tree classifier. We defined the hyperparameter grid to include different values of `max_depth`, `min_samples_split`, and `min_samples_leaf`. We then split the data into training and testing sets using `train_test_split`, create a decision tree classifier object, and create a grid search object with `GridSearchCV`. We fit the grid search object to the training data using the `fit` method, and finally print the best hyperparameters and best score. The best score we got was 68.28%.

b. Random Forest Classifier

After implementing Random Forest Classifier to predict the right career stream, we found that the classifier accuracy was coming out to be 84.17%. We also took manual inputs and got accuracy of 71.42%.

Optimizing Random Forest Performance: Exploring Hyperparameter Tuning Techniques

We have used Grid Search to tune the hyperparameters. We define the hyperparameter grid to include different values of `n_estimators`, `max_depth`, `min_samples_split`, and `min_samples_leaf`. We then split the data into training and testing sets using `train_test_split`, create a random forest classifier object, and create a grid search object with `GridSearchCV`. We fit the grid search object to the training data using the `fit` method, and finally print the best hyperparameters and best score.

The best score found out to be 85.68%.

c. SVM

After implementing SVM Classifier to predict the right career stream, we found that the classifier accuracy was coming out to be 75.81%. We also took manual inputs and got accuracy of 71.42%.

Optimizing SVM Performance: Exploring Hyperparameter Tuning Techniques:

We have used Grid Search to tune the hyperparameters. We define the hyperparameter grid to include different values of C, kernel, degree, and gamma. We then split the data into training and testing sets using `train_test_split`, create an SVM object, and create a grid search object with `GridSearchCV`. We fit the grid search object to the training data using the `fit` method, and finally print the best hyperparameters and best score. The best score found out to be 77.19%.

d. Naïve Bayes

After implementing Naïve Bayes Classifier to predict the right career stream, we found that the classifier accuracy was coming out to be 69.93%. We also took manual inputs and got accuracy of 71.42%.

Optimizing Naïve Bayes Performance: Exploring Hyperparameter Tuning Techniques:

In scikit-learn, the Gaussian Naïve Bayes implementation has one hyperparameter. The hyperparameter used is `var_smoothing`, which is a smoothing parameter for variance in Gaussian Naïve Bayes. We have set the hyperparameters for the Gaussian Naïve Bayes classifier using the `set_params()` method, passing the hyperparameters dictionary. Then we fitted the classifier to the training data using the `fit()` method and then making predictions on the testing data using the `predict()` method and storing the predicted labels in `y_pred` and eventually predicting the best score. The best score came out to be 72.07%.

e. KNN

After implementing KNN Classifier to predict the right career stream, we found that the classifier accuracy was coming out to be 70.84%. We also took manual inputs and got accuracy of 71.42%.

Optimizing KNN Performance: Exploring Hyperparameter Tuning Techniques:

The hyperparameters used for the K-nearest neighbors (KNN) classifier are “`n_neighbors`”: Number of neighbors to consider. “`weights`”: Weight function used in prediction. “`p`”: Power parameter for Minkowski distance. We have first split the data into training and testing sets, Create and fit the KNN classifier, set hyperparameters, A dictionary ‘hyperparameters’ is defined to store the hyperparameter values for the KNN classifier, Pass the hyperparameters to the classifier, Fit the model to the training data and then predicted the best score. The Best Score came out to be 71.88% (Table 7).

6 Limitations and Challenges

While machine learning techniques can provide valuable insights and assist in career counseling, there are several limitations to consider:

- **Lack of human intuition and empathy:** Machine learning algorithms rely on data patterns and statistical analysis, but they may not capture the nuanced human aspects

Table 7 Accuracy comparison

Classification algorithm	Accuracy obtained (before hyperparameter)	Accuracy obtained (after hyperparameter)
Decision tree	66.33	68.28
Random forest	84.17	85.68
Support vector classification	75.81	77.19
Naïve Bayes	69.93	72.07
K-nearest neighbors	70.84	71.88

required for effective career counseling. Human intuition, empathy, and understanding of individual circumstances are often crucial in career decision-making.

- **Limited input data:** Machine learning algorithms depend on the quality and quantity of input data. Career counseling involves complex and multifaceted factors, including personal interests, values, skills, and aspirations, which may not be easily quantifiable or available in structured datasets. This limitation can lead to oversimplification or inadequate recommendations.
- **Changing job market dynamics:** The job market is dynamic and constantly evolving, influenced by economic trends, technological advancements, and societal changes. Machine learning models trained on historical data may not effectively adapt to these shifts or accurately predict future job market demands.
- **Lack of context and cultural sensitivity:** Machine learning models are typically trained on diverse datasets, but they may not capture the specific cultural, social, or contextual factors that can significantly impact career choices. Career counseling requires considering individual backgrounds, cultural influences, and local job markets, which can be challenging for machine learning algorithms.
- **Ethical concerns and bias:** Machine learning models can perpetuate biases present in the data they are trained on, potentially leading to discriminatory or unfair outcomes in career counseling. Biases related to gender, race, socioeconomic status, or other protected characteristics can inadvertently influence recommendations and hinder equal opportunities.
- **Limited transparency and interpretability:** Many machine learning techniques, such as deep learning models, are often considered black boxes, making it challenging to understand how they arrive at specific recommendations. This lack of transparency can be problematic when providing career advice, as it becomes difficult to explain the reasoning behind the recommendations to individuals seeking guidance.
- **Personal growth and self-discovery:** Career counseling is not solely about finding the most statistically optimal career path. It also involves personal growth, self-discovery, and exploration of individual interests and passions. Machine learning techniques may not adequately account for these subjective and non-quantifiable aspects of career decision-making.

It's important to strike a balance between leveraging machine learning techniques for data-driven insights and incorporating the expertise of human career counselors to overcome these limitations and provide holistic career guidance.

7 Ablations or Drawback of the Research

The application of machine learning techniques in career counseling can have several benefits and opportunities for improvement. However, it's important to consider potential ablations or drawbacks that may arise when employing these techniques. Here are some factors to consider:

- **Limited human interaction:** Machine learning techniques can automate certain aspects of career counseling, which may reduce the level of human interaction. While this can be efficient and cost-effective, it may also lead to a less personalized experience for individuals seeking career guidance. Some individuals may prefer face-to-face interactions and value the empathy and understanding that human counselors provide.
- **Bias and fairness:** Machine learning models are trained on existing data, and if the training data contains biases, those biases can be perpetuated and amplified in the model's predictions. This can lead to unfair or discriminatory recommendations in career counseling. For example, if historical data shows a bias towards certain demographics or industries, the model may unknowingly recommend similar paths to individuals without considering their unique skills, interests, or potential.
- **Lack of context and nuance:** Career counseling involves understanding a person's unique circumstances, aspirations, and values. Machine learning models typically analyze large amounts of data but may struggle to capture the full context and nuances of an individual's career situation. Factors such as personal circumstances, work-life balance, and cultural aspects might not be adequately addressed by machine learning algorithms alone.
- **Limited understanding of emerging careers:** The job market is constantly evolving, and new career paths and industries emerge regularly. Machine learning models are trained on historical data, and if the data does not encompass the latest trends and emerging fields, the recommendations may not be up-to-date or relevant. Human career counselors, on the other hand, can stay current with industry trends and adapt their guidance accordingly.
- **Lack of transparency:** Machine learning models can sometimes be considered as black boxes, meaning that their decision-making process is not easily interpretable or explainable. This lack of transparency can make it challenging for individuals to understand the rationale behind the recommendations provided by the model. Transparent explanations and justifications are important for individuals to feel confident in the guidance they receive.

To address these ablations, a balanced approach can be adopted by combining the strengths of machine learning techniques with the expertise and empathy of human career counselors. This hybrid model can leverage the efficiency of automated systems while ensuring personalized, fair, and contextually relevant career advice. Human counselors can collaborate with machine learning models, providing oversight, interpreting recommendations, and tailoring them to the individual's unique needs.

8 Conclusion and Future Line of Work

In this research, various machine learning algorithms are applied to the dataset to predict the right career option. The different algorithms were Decision Tree, Random Forest, SVM, Naïve Bayes, and KNN. Out of these, Random Forest gave the highest accuracy of

84.17%, and after hypertuning, it gave the highest accuracy of 85.68. We also gave some manual inputs to the system and found out that the Random Forest gave the highest accuracy of 85.71%. Further, the future scope of career prediction using machine learning is promising, with potential for enhanced accuracy, personalised recommendations, real-time labour market analysis, career pathway prediction, incorporation of emerging technologies, and ethical considerations. Continued research and advancements in machine learning techniques have the potential to revolutionise career planning and decision-making, helping individuals make informed choices about their professional futures.

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Code Availability Code for implementations are available on request due to privacy or other restrictions.

Declarations

Conflicts of interest The authors declare that there is no conflict of interest.

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